

Geoffrey Lazer

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Portfolio: Unity/XR | Available April 2027 | Open to roles across Canada

Professional Summary

Unity/XR software engineer with an MSc in Computing Science. Builds Quest research prototypes, medical visualization and AI-enabled interfaces, with spatial interaction, device validation and documented deployment and handoff workflows.

Technical Skills

XR / real-time 3D: Unity, C#, OpenXR, Meta Quest 3, spatial UI, hand/controller interaction

Imaging / data: DICOM, NIFTI, VTK, C++, Python, NumPy, QGIS, GeoJSON, REST APIs

Engineering: Git, GitHub Actions, automated validation, Android builds, CSV logging

Professional Experience

XpertVR

Ontario

Unity VR / AI Developer - Fixed-term contract

Oct 2026 – Present

- Current R&D scope: connect AI/LLM communication with live Unity simulation state and translate participant instructions into bounded tasks for VR training.
- Responsibilities include AI guardrails, validation, NPC/speech integration, response-latency optimization and VR performance testing.

University of Alberta

Edmonton, Alberta

XR/ML Engineer

May 2026 – Present

- For ergonomic review, built a Unity/OpenXR Quest 3 workspace combining three synchronized body views, risk highlighting, playback and spoken guidance.
- Integrated authenticated RunPod processing and package imports with a 50-slot session manager, enabling participant-specific coaching without rebuilding the APK.
- Resolved input, download-recovery and playback issues; the prototype passed 15 automated checks and averaged 72.1 FPS in a representative Quest 3 validation.
- Translated a construction-safety protocol into six five-minute sessions and 120 trials per participant, integrating MaxJS and LinRL, 12 soundtracks and reproducible scheduling for a planned 30-participant study.
- Delivered CSV exports, signed Quest builds and maintenance guides for research handoff; a validated update preserved all 138 existing headset research files.

FireSafe AI

Edmonton, Alberta

Machine Learning Engineer

Oct 2025 – May 2026

- For wildfire modeling, built Python ETL combining Canadian/U.S. events, weather and geospatial data into standardized 15-day sequences, with reusable classification and regression, benchmarking and inference workflows documented for handoff.

Luxsonic Technologies Inc.

Saskatoon, Saskatchewan

Software Developer (Internship)

May 2025 – Aug 2025

- For immersive scan inspection, developed Unity/C# DICOM volume rendering, slicing and transfer functions, enabling interactive cross-sectional review.
- Built an OpenXR gesture keyboard and integrated offline Vosk speech-to-text, enabling gesture and voice input in VR.
- Explored Dicom2Mesh/C++/VTK mesh extraction to evaluate lighter real-time medical-visualization paths.

University of Alberta

Edmonton, Alberta

Graduate Teaching Assistant, CMPUT 402

Jan 2025 – Apr 2025

- Guided CMPUT 402 teams in CI/CD and automated testing with GitHub Actions, JUnit, SonarQube and mutation testing, providing actionable software-quality feedback.

Selected XR & Interactive Projects

Volume Rendering and 3D Slicing for Medical Imaging

Jan 2025 – Apr 2025

- To inspect scan structures spatially, implemented real-time NIFTI/DICOM volume rendering and interactive box-cutout slicing in Unity/C#, delivering an XR research prototype.

Demo: youtu.be/wo76y5mV-r4

Agricultural Terrain Visualization

Sep 2024 – Dec 2024

- Prepared QGIS/GeoJSON data, imported terrain into Unity and added automatic pathfinding, delivering a Quest 3 demonstration for immersive agricultural-plot navigation.

Demo: youtube.com/shorts/IbXerF5edYM

Helex VR

Dec 2021 – Aug 2022

- Contributed Unity VR training and YOLO exercise feedback; the team won Smart India Hackathon 2022.

Education

University of Alberta

Edmonton, Alberta

MSc, Computing Science

Sep 2024 – Jul 2026

Loyola-ICAM College of Engineering and Technology

Chennai, India

BE, Computer Science and Engineering

Jul 2020 – Jun 2024